Reinforcement learning versus an optimized forecast rule and an optimized fixed orifice for the underdrain valve of a bioretention cell

St. Francis cell, conventional (30.2 mm) footprint, perfect rainfall forecast

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Abstract

We compare three ways of operating the underdrain valve of a bioretention cell: a fixed orifice of the best size, the forecast-informed rule of Gabbani et al. with its three parameters optimized, and a reinforcement-learning (RL) policy trained with proximal policy optimization (PPO) as an hourly correction on top of that rule. All three are optimized for the same objective, using the same OpenHydroQual model, rainfall record (2020–2023) and perfect forecast. Peak flow is measured as the outflow volume above a preset limit, $\int \max(Q - Q_{\lim}, 0)\, dt$ with $Q_{\lim} = 50\,\mathrm{m^3/d}$, which is additive in time, unlike the upper-tail statistic of the earlier study. Two objectives are considered: the excess volume alone (case 1), and the excess volume plus the young-water load of an age tracker, normalized by the passive cell and weighted equally (case 2). In case 1 the optimized rule ($J = 0.0768$) beats the best fixed orifice (0.0889) by 14%, and PPO does not improve on the rule: after 3 million training steps the final policy scores 0.0831, and no trained checkpoint beats the rule. In case 2 the rule ($J = 0.285$) beats the orifice (0.426) by 33%, and the final PPO policy beats the rule by a further 10% ($J = 0.256$; checkpoints late in training reach 0.186): it keeps the cell three times fuller on average, cutting the young-water load by 43%, at the cost of 60% more excess volume. These are in-sample results. Trained on 2020–2022 and tested on 2023, a dry year with no flow above the limit, the case-2 policy keeps a third of the rule's young-water load. With 2020 held out (rule and orifice re-tuned and PPO retrained on 2021–2023), the case-2 gain holds on the unseen year: PPO scores $J = 0.447$ on 2020 against 0.673 for the rule (34% lower) and 0.894 for the orifice, lowering both the excess and the young-water load. In case 1 PPO is again slightly worse than the rule on 2020 (0.200 against 0.189).

1 Introduction

Forecast-informed real-time control of a bioretention underdrain can lower outflow peaks by releasing water ahead of a storm and holding it otherwise. Gabbani et al. proposed a three-parameter rule: the valve opens in proportion to the square root of a kernel-smoothed rainfall forecast and closes when that forecast is below a threshold. The later forecast-uncertainty study found that, once the overflow weir is realistic and the time step converged, this rule beats a well-sized fixed orifice only modestly on peak flow, but by 30–40% when residence time is also rewarded.

Here we ask whether a learned policy can do better than the rule when both have the same information (a perfect forecast) and the same objective. To make the comparison fair, the rule and the fixed orifice are both re-optimized on the objective used to train the RL policy, and all three are scored on the same whole-record runs.

2 System and model

The model is the St. Francis bioretention cell of the forecast-uncertainty study at half of its as-built plan area: 30.2 mm of storage over the 7300 m^2 catchment, the conventional water-quality sizing, where that study found forecast control most valuable. Runoff from four catchment segments enters a pond (crest 158.2 m, 1 m sharp-crested weir to the catch basin), infiltrates through the engineered media into an aggregate storage layer, and leaves either to the native soil or through the controlled underdrain (diameter 0.2 m) to the catch basin. The catch-basin outflow pipe, carrying both the weir overflow and the underdrain flow, is where all objectives are measured. Forcing is the hourly 2020–2023 record (precipitation, and temperature, humidity, radiation and wind for Penman evapotranspiration). A Laplace age tracker (unit concentration on every inflow, first-order decay $\lambda = 1/\mathrm{d}$) carries $E[e^{-\lambda\,\mathrm{age}}]$, so its exported load is the *young-water load*: lower means water is held longer.

The model files are the study's residence-time models (`rt_campaign/models/R1_f0.50_half_30.2mm/sigma_0_ts_2`) and its passive reference, with only three changes: the objectives are replaced (Section 3), the templates are read from the current OpenHydroQual master (whose changes since the pinned commit only add new types), and, for the fixed orifice and the RL policy, the valve is replaced by a controlled orifice (below). Numerics are the study's: initial step 0.01 d, maximum step 0.1 d.

3 Objectives

Both objectives are integrals over the record, so they are additive in time and can serve directly as an RL reward:

$$\mathrm{Excess} = \int_{t_0}^{t_1} \max\big(Q(t) - Q_{\mathrm{lim}},\, 0\big)\, dt, \qquad Q_{\mathrm{lim}} = 50\,\mathrm{m}^3/\mathrm{d}\ (0.58\,\mathrm{L/s}), \tag{1}$$

$$\mathrm{Age} = \int_{t_0}^{t_1} Q(t)\, c_{\mathrm{age}}(t)\, dt, \tag{2}$$

where Q is the catch-basin outflow and c_{age} the age-tracer concentration. The limit was chosen between the rule's typical release rate (50–80 m^3/d) and the passive cell's 99th-percentile outflow (about 100 m^3/d). Each term is normalized by the passive cell (valve replaced by a free-draining underdrain): $\mathrm{Excess}_p = 2793\,\mathrm{m}^3$, $\mathrm{Age}_p = 2753$. The two cases are

$$\text{case 1:}\quad J_1 = \mathrm{Excess}/\mathrm{Excess}_p, \tag{3}$$

$$\text{case 2:}\quad J_2 = \mathrm{Excess}/\mathrm{Excess}_p + \mathrm{Age}/\mathrm{Age}_p, \tag{4}$$

so the passive cell scores 1 and 2. Lower is better.

4 Controllers

Fixed orifice. The controlled-orifice link has the same head dependence as the valve, $Q_v = 86400\,u\,A\sqrt{2g}\,\Delta h/\sqrt{\Delta h + 1\,\mathrm{mm}}$, with A the valve area and u the opening. The best constant u is found by a sweep of 48 sizes (geometric, 2×10^{-4} to 0.05) for each case.

Forecast rule. With a perfect forecast the rule is

$$u_{\text{rule}}(t) = \begin{cases} \sqrt{\hat{I}(t)/S} & \hat{I}(t) > I_0 \\ 0 & \text{otherwise,} \end{cases} \tag{5}$$

where $\hat{I}$ is the rainfall rate r convolved with a truncated, forward-looking exponential kernel:

$$\hat{I}(t) = \int_0^{2\tau_k} K(\tau)\, r(t+\tau)\, d\tau = (\tilde{K} * r)(t), \qquad K(\tau) = \lambda_k\, e^{-\lambda_k \tau}, \quad \lambda_k = 1/\tau_k, \tag{6}$$

with $\tilde{K}(\tau) = K(-\tau)$, so the convolution looks ahead in time: with a perfect forecast, $\hat{I}(t)$ weighs the rain of the next $2\tau_k$ days, most heavily the rain closest to t. The kernel is truncated at two time constants, where it keeps $1 - e^{-2} \approx 86\%$ of its weight. OpenHydroQual evaluates (6) on the rainfall series (t_i, r_i) by the trapezoidal rule, $\hat{I}(t) = \sum_i r_i\, \lambda_k\, \frac{1}{2}\big(e^{-\lambda_k(t_i - t)} + e^{-\lambda_k(t_{i+1} - t)}\big)(t_{i+1} - t_i)$, and the controller here uses this sum exactly as the model does.[1] The three parameters (S, I_0, τ_k) are optimized by CMA-ES in log space (population 12, 35 generations, bounds $S \in [5, 2000]$, $I_0 \in [5 \times 10^{-7}, 0.015]$ m/d, $\tau_k \in [0.001, 100]$ d), separately for each case, starting from the better of the study's two perfect-forecast optima. Re-tuning lowered J by 34% (case 1) and 23% (case 2) relative to those starting points, so the study's parameters, tuned for a different peak statistic, would have been a weak benchmark. The rule is scored in its own model, where the valve responds to $\hat{I}$ continuously.

PPO. The RL policy corrects the rule every hour: $u = \text{clip}(u_{\text{rule}} + a\, u_s,\ 0,\ 0.03)$ with $a \in [-1, 1]$ and $u_s = 0.004$, so $a = 0$ is the rule applied hourly. The observation (16 values) holds the ponded, aggregate-layer, media and catch-basin storages, the outflow and weir flow, the rule's opening, the perfect-forecast mean rainfall over the next 0–3 h, 3–12 h, 12–24 h, 1–3 d and 3–7 d, the past day's rainfall, $\hat{I}$, and the season. The reward is minus the increase of J over the hour, so the return of an episode is minus its J. Episodes are 30-day windows starting at random times in the record, each initialized from the state of a continuous run of the hourly rule at that time. Training: actor-critic with two hidden layers of 64 (tanh), PPO with GAE ($\gamma = 0.995$, $\lambda = 0.95$, clip 0.2, 10 epochs, minibatch 1024, learning rate 3×10^{-4}), 6 workers × 720 steps per update (4,320 steps; 694 updates, 3 million steps, one seed per case), reward normalization, and the actor first cloned to $a = 0$ (the rule) with exploration $\log \sigma = -1.2$. Checkpoints every 20 updates were scored deterministically over the whole record. The policy reported is the final one; the checkpoints show how much the result depends on where training stops.

5 Simulation and verification

Each controller is simulated with a compiled OpenHydroQual kernel (`ohq_generate`), driven from Python: the opening is a kernel parameter set before each hour, and the objectives are integrated from the kernel's per-step series as OpenHydroQual integrates them. A whole-record run takes about 12 s (more than 18 minutes interpreted), and a 30-day training episode 0.4 s. Three checks:

- The hourly rule on the orifice model reproduces the rule model to within 0.2% (case 1) and 0.6% (case 2) in J; the remaining difference is that the native valve switches within the hour.

[1] Versions of OpenHydroQual before 2 November 2026 omitted the $\frac{1}{2}$, doubling $\hat{I}$. Because $\hat{I}$ enters the rule only through $\hat{I}/S$ and the comparison with I_0, this is equivalent to halving S and I_0: the same controller, the same J. All parameters here are given for the corrected kernel; parameters tuned with the earlier version convert as $S \to S/2$, $I_0 \to I_0/2$.

Table 1: Whole-record (2020–2023) results with a perfect forecast. Excess in m^3; peak outflow in m^3/d; hours with outflow above $Q_{\lim}$; weir overflow volume in m^3. PPO is the final policy (update 694) of each training run.

	J	Excess	Excess/Ex$_p$	Age/Age$_p$	Peak	Hours $> Q_{\lim}$	Weir vol.
Case 1: excess only							
Passive (free draining)	1.000	2793	1.000	1.000	755	1156	0
Best fixed orifice	0.089	248	0.089	0.451	526	80	259
Tuned forecast rule	0.077	215	0.077	0.327	266	195	140
PPO (final policy)	0.083	232	0.083	0.243	374	180	357
Case 2: excess + residence time							
Passive (free draining)	2.000	2793	1.000	1.000	755	1156	0
Best fixed orifice	0.426	364	0.130	0.295	632	135	570
Tuned forecast rule	0.285	255	0.091	0.194	489	214	502
PPO (final policy)	0.256	407	0.146	0.111	354	262	754

Table 2: Optimized parameters.

		Forecast rule		Fixed orifice
	S	I_0 (m/d)	τ_k (d)	opening u
Case 1	272	1.6×10^{-6}	4.49	0.00377
Case 2	804	5.0×10^{-7} (bound)	1.95	0.00210

- An episode started from a stored state reproduces the continuous run over the same window to 0.01%.
- Storages of the compiled kernel match the interpreted model to 0.1% over a 30-day test with a switching opening.

6 Results

Case 1: excess only. The optimized rule beats the best fixed orifice by 14% (Table 1, Figure 1). It halves the peak outflow (266 against 526 m^3/d) and the weir overflow (140 against 259 m^3), although it exceeds the limit for more hours (195 against 80): it releases more, at moderate rates, before storms. PPO does not improve on it. After an early excursion (up to 0.233 at update 60) training settles close to the rule, between 0.077 and 0.084 from update 300 on (Figure 2), but no trained checkpoint beats the starting policy, the hourly rule ($J = 0.0767$). The final policy scores 0.0831, 8% worse than the rule. It holds more water than the rule (14.9 m^3 ponded on average against 5.2 m^3, valve shut 24% of the time), which lowers the hours above the limit (180 against 195) and, as a side effect, the young-water load (0.243 against 0.327), but raises the peak (374 against 266 m^3/d) and the weir overflow (357 against 140 m^3).

Case 2: excess plus residence time. The rule beats the fixed orifice by 33%: it is lower on both terms (excess 0.091 against 0.130, young-water load 0.194 against 0.295 of passive), which no single orifice size achieves (Figure 3; the orifice sizes trace a curve that the rule lies well inside). The final PPO policy beats the rule by a further 10% ($J = 0.256$). It reaches this by a different balance (Figure 3): the young-water load falls by 43% (0.111 against 0.194 of passive) while the excess rises by 60% (0.146 against 0.091). The policy keeps the valve shut 37% of the time (the

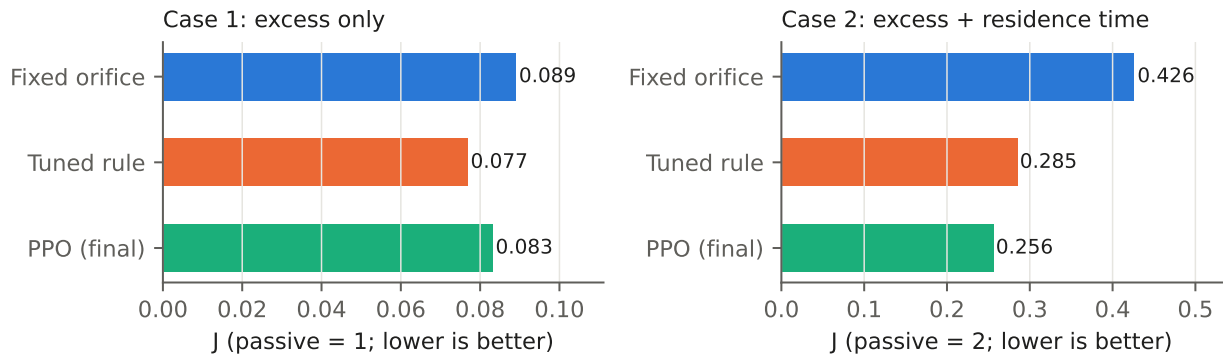


Figure 1: Objective J of the three optimized controllers over the whole record.

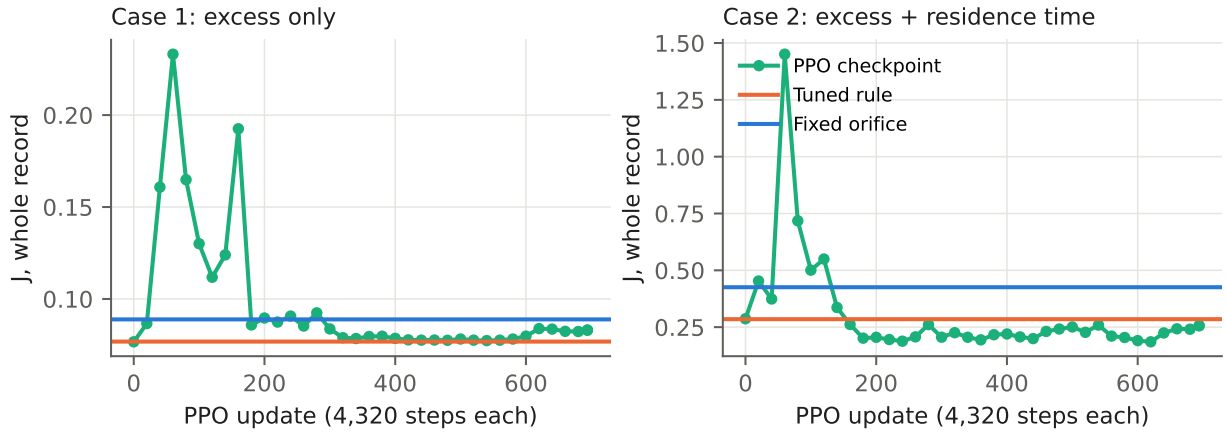


Figure 2: Whole-record J of the PPO checkpoints (update 0 is the rule applied hourly). The last point is the final policy (update 694, 3 million steps).

hourly rule 26%), keeps on average 58 m^3 ponded against 17 m^3, and so overflows the weir more (754 against 502 m^3) and exceeds the limit for more hours (262 against 214), while lowering the peak outflow (354 against 489 m^3/d).

Where training stops matters (Figure 2). The first 140 updates are worse than the rule, up to $J = 1.45$ at update 60, worse than the passive cell, as the policy leaves the rule. From update 160 on, all 28 checkpoints are better than the rule, with a median of 0.214 (25% below the rule) and a range of 0.186 to 0.262. The best, update 620 ($J = 0.186$, 35% below the rule), is lower than the rule on both terms (excess 0.103, young-water load 0.083), so a better balance than the final policy's exists. The final policy is near the upper end of this range; training does not converge to a single policy, and the gain at any one stopping point varies between about 10% and 35%.

The largest storm. In the largest 24-hour storm of the record (Figure 5), all controlled cells fill to about 200 m^3 and overflow; the differences are before and after the storm. The rule and the case-2 PPO policy open ahead of the storm as the forecast rises, which the fixed orifice cannot; PPO opens slightly less (mean opening 5.1×10^{-3} over the three days before the storm against 5.8×10^{-3}). As the storm peaks, PPO almost closes the valve (opening 0.4×10^{-3}, against the rule's minimum of 1.5×10^{-3}), and then, over the following days, opens it wider than the rule (mean 3.5×10^{-3} against 3.0×10^{-3}), so the cell is drained further by the end of the window (134

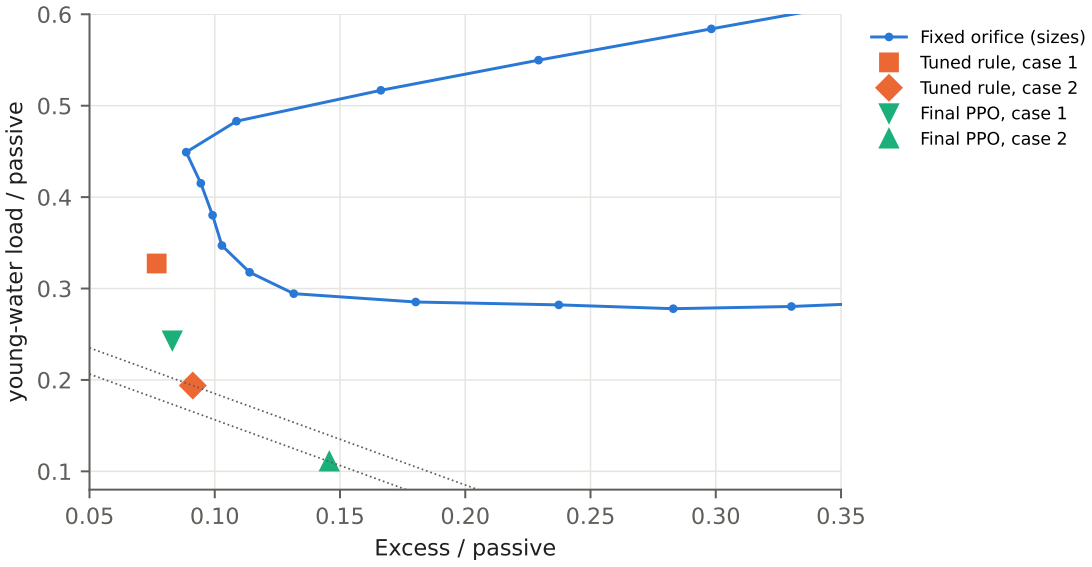


Figure 3: Excess and young-water load (both relative to passive). The fixed orifice traces a curve as its size varies; dotted lines are lines of equal case-2 J through the rule and the final PPO policy (case 2).

against $152\,\mathrm{m}^3$). In this storm PPO holds water at the peak and releases it afterwards.

6.1 Held-out year: 2023

The first test held out the last year: the passive reference, orifice sweep, rule tuning, snapshots and PPO training were redone on 2020–2022, and each controller was scored on 2023 (same protocol as for 2020 below). The re-tuned rule and orifice are essentially those of the whole record (case 1: $S = 273$, $I_0 = 6.6 \times 10^{-6}$, $\tau_k = 4.48$, orifice 0.00377; case 2: $S = 800$, $I_0 = 5.0 \times 10^{-7}$, $\tau_k = 1.96$, orifice 0.00210), since 2023 adds no excess. 2023 is dry: the passive cell exceeds the limit by only $346\,\mathrm{m}^3$, and no controlled cell exceeds it at all. In case 1 every controller therefore scores $J = 0$ on 2023 and the test cannot separate them. In case 2 only the young-water load is left: the final PPO policy scores 0.051 against 0.155 for the rule and 0.293 for the orifice, a third of the rule's load, consistent with its in-sample behaviour of holding water longer. On the training years it scores 0.249 against the rule's 0.305. A year with large storms is needed to test the excess term out of sample, hence the 2020 test.

6.2 Held-out year: 2020

To test whether the gains carry over to storms the controllers have not seen, 2020, the wettest year, was held out. Everything was redone on 2021–2023 alone: the passive reference, the orifice sweep, the CMA-ES tuning of the rule, the snapshots and the PPO training (episodes start only in 2021–2023; 3 million steps, 694 updates of 6 workers $\times$ 720 steps, one seed). Each controller is then run over the whole 2020–2023 record under its own control, and the integrals are split at 1 January 2021, so the 2020 score starts from the state the controller itself produced. Both windows are normalized by the passive cell in the same window (2020: $\mathrm{Excess}_p = 1177\,\mathrm{m}^3$, $\mathrm{Age}_p = 990$; 2021–2023: $1617\,\mathrm{m}^3$ and 1763). The PPO policy reported is the final one, chosen without looking at 2020. The re-tuned parameters are $(S, I_0, \tau_k) = (398,\ 5.0 \times 10^{-7}$ (bound), $3.00)$ and orifice opening 0.00236 for case 1, and $(1470,\ 1.6 \times 10^{-6},\ 3.47)$ and 0.00166 for case 2.

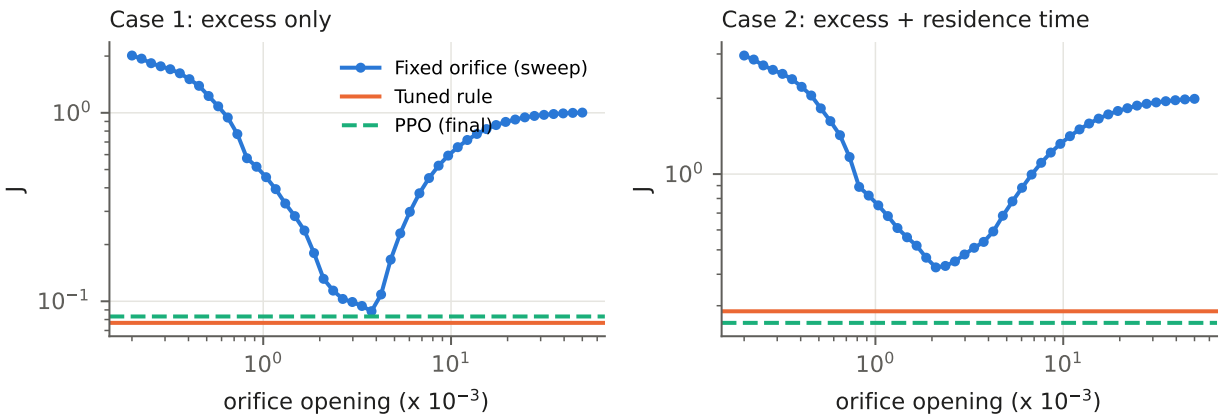


Figure 4: J of the fixed orifice against its opening, with the optimized rule and the final PPO policy.

Table 3: Held-out year. Controllers tuned or trained on 2021–2023, scored on 2021–2023 and on 2020, each relative to the passive cell in that window. Excess in m^3. PPO is the final policy (update 694). In case 1 Age/Age$_p$ is not part of J and is shown for reference.

	J, 2021–23	J, 2020	Excess, 2020	Ex/Ex$_p$, 2020	Age/Age$_p$, 2020
Case 1: excess only					
Best fixed orifice	0.000	0.264	311	0.264	0.343
Tuned forecast rule	0.000	0.189	222	0.189	0.308
PPO (final)	0.005	0.200	236	0.200	0.408
Case 2: excess + residence time					
Best fixed orifice	0.247	0.894	595	0.506	0.389
Tuned forecast rule	0.130	0.673	438	0.373	0.301
PPO (final)	0.049	0.447	311	0.265	0.182

2020 is the wettest year of the record: it holds 42% of the passive excess volume of the four years. 2021–2023 has almost no flow above the limit once the cell is controlled. In case 1 the rule and the orifice have no excess at all on the training years ($J = 0$), so there is nothing for PPO to improve and nothing to choose between the two benchmarks there.

Case 2. The ranking of the in-sample results holds on 2020 (Table 3). The rule beats the orifice by 25% (0.673 against 0.894), and the final PPO policy beats the rule by 34% ($J = 0.447$). Unlike the in-sample policy, it does so on both terms: 29% less excess volume (311 against 438 m^3) and 39% less young-water load (0.182 against 0.301 of passive). The gain is not a matter of picking a lucky checkpoint (Figure 6). After a transient in the first 40 updates, when the policy first leaves the rule and is worse than the orifice, every one of the 32 checkpoints from update 80 to 694 is better than the rule on 2020, between 0.426 and 0.609; the checkpoint best on the training years (update 320, $J = 0.045$) scores 0.440 on 2020, close to the final policy. The gain is smaller on 2020 than on the training years (63%), as expected out of sample, but it is larger than the 10% gain of the final in-sample policy, and within the 10–35% range of the in-sample checkpoints.

Case 1. PPO does not beat the rule on 2020 either: the final policy scores 0.200 against 0.189 (6% worse), still well ahead of the orifice (0.264). Since the training years give it almost no excess to reduce, the policy drifts towards holding water (its young-water load is 0.408, against 0.308 for

the rule), which costs it a little excess in 2020. No case-1 checkpoint after update 0 beats the rule on 2020.

7 Discussion and limitations

The results repeat the pattern of the forecast-uncertainty study. When only high flows count, the optimized rule already captures most of what the forecast offers, and PPO finds nothing better. When residence time also counts, there is a balance between holding and releasing water that a rule driven only by the forecast does not express well, and PPO, which also sees the storages, finds a better one. The case-2 result rests on the following qualifications.

1. **In-sample.** As in the study's GA campaign, all controllers are tuned and scored on the same 2020–2023 record, and a gain from a 4,500-parameter network may partly reflect the particular storms of the record. The hold-out tests answer this in part: trained and tuned without 2020, PPO beats the rule on 2020 by 34% in case 2 (Section 6.2), and trained without 2023 it keeps a third of the rule's young-water load in that dry year (Section 6.1). Each is one year and one seed, however, and the training years contain almost no flow above the limit under control, so the excess term contributes little to the training signal; a held-out year with fewer large storms might not show the same gain.

2. **Single, unconverged training runs.** Each policy is one seed of 3 million steps. The case-2 checkpoints keep moving between $J = 0.186$ and 0.262 late in training, so the gain over the rule depends on where training stops (10% for the final policy, 35% for the best checkpoint), and the final policy is not the best one. Repeated seeds, and a stopping rule chosen on a validation period, would make the reported gain less arbitrary.

3. **Rule optimum.** Each rule was tuned by one CMA-ES run, and in case 2 the threshold sits on its lower bound. Restarts and a wider bound would confirm that the rule is at its best within its form.

4. **Perfect forecast and hourly decisions.** With forecast error the rule's advantage over the orifice shrinks, especially in larger cells; how a learned policy degrades is untested. The policy acts hourly, while the native rule responds continuously (which costs it nothing here: the hourly rule scores within 0.6%).

5. **The limit** Q_{lim} was set at $50\,\mathrm{m}^3/\mathrm{d}$; other limits, or a nonlinear function of the excess, weight the peaks differently and may change the ranking.

8 Conclusions

- With the excess volume above $50\,\mathrm{m}^3/\mathrm{d}$ as the objective, the optimized forecast rule beats the best fixed orifice by 14%, and PPO cannot improve on the rule, in sample or on the held-out year 2020.

- With excess volume and residence time weighted equally, the rule beats the orifice by 33%, and PPO beats the rule by a further 10% in sample with its final policy (25% at the median late checkpoint), mainly by holding water longer.

- The case-2 gain holds out of sample: trained and tuned on 2021–2023, PPO beats the rule by 34% on the held-out year 2020, lowering both the excess volume and the young-water load;

on the dry held-out year 2023 it keeps a third of the rule's young-water load. Repeated seeds and a principled stopping rule are still needed.

A Reproducing

All code is in `OpenHydroQual-RL/stfrancis`: `build_models.py` (models), kernels via `ohq_generate`, `tune_rule.py` (passive reference, orifice sweep, CMA-ES rule tuning), `snapshots.py`, `sf_env.py` (environment), `../train_ppo.py --env stfrancis`, `evaluate_sf.py` and `make_paper.py`; `run_rl.sh` chains the RL steps. The hold-out test is `campaign_split.sh ho2020` (tuning, snapshots, training and scoring on 2021–2023 with 2020 held out) and `make_holdout.py` (Figure 6).

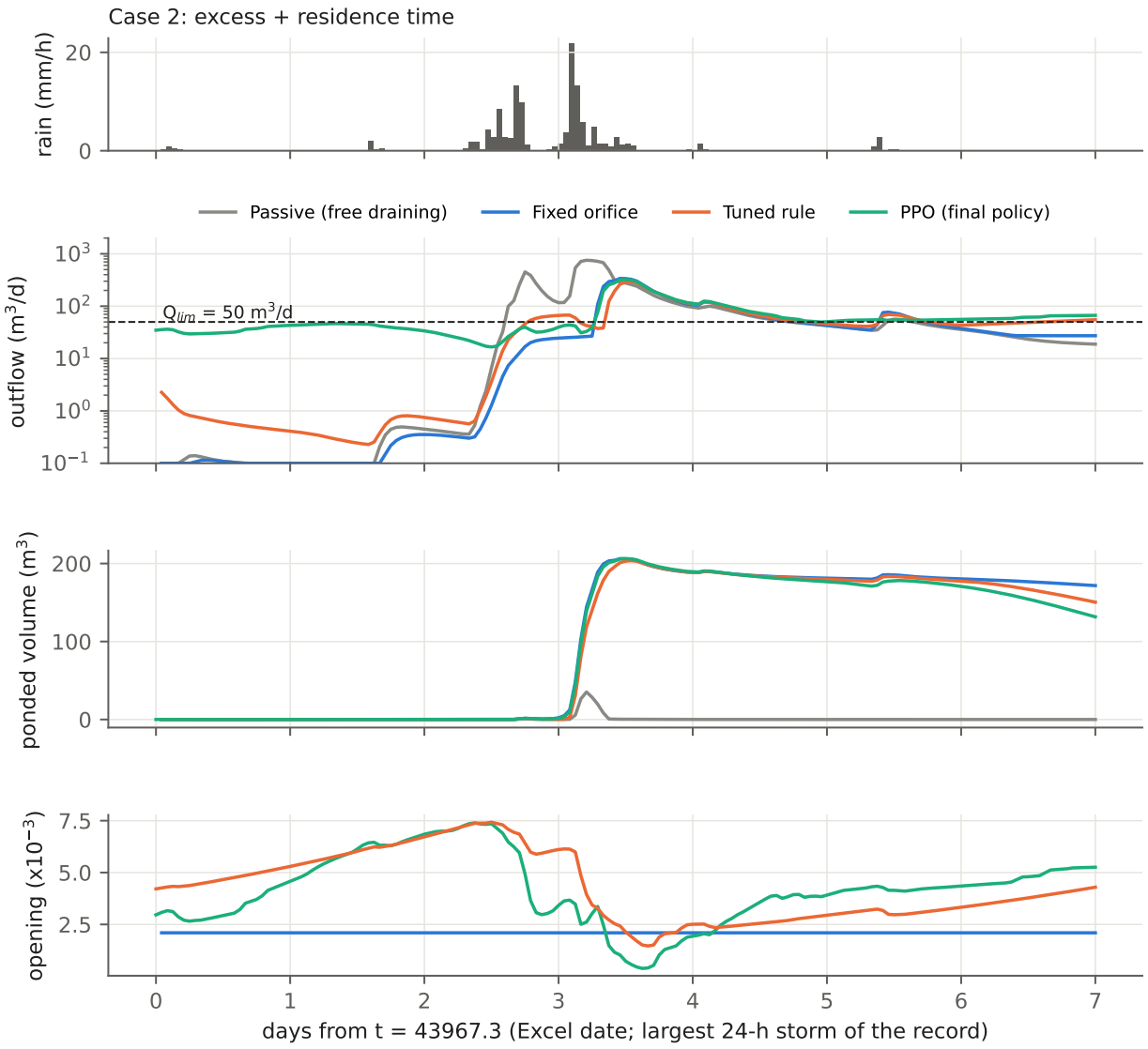


Figure 5: Largest 24-hour storm, case-2 controllers: rainfall, outflow (log scale, with Q_{lim}), ponded volume and valve opening (the rule's opening is shown applied hourly).

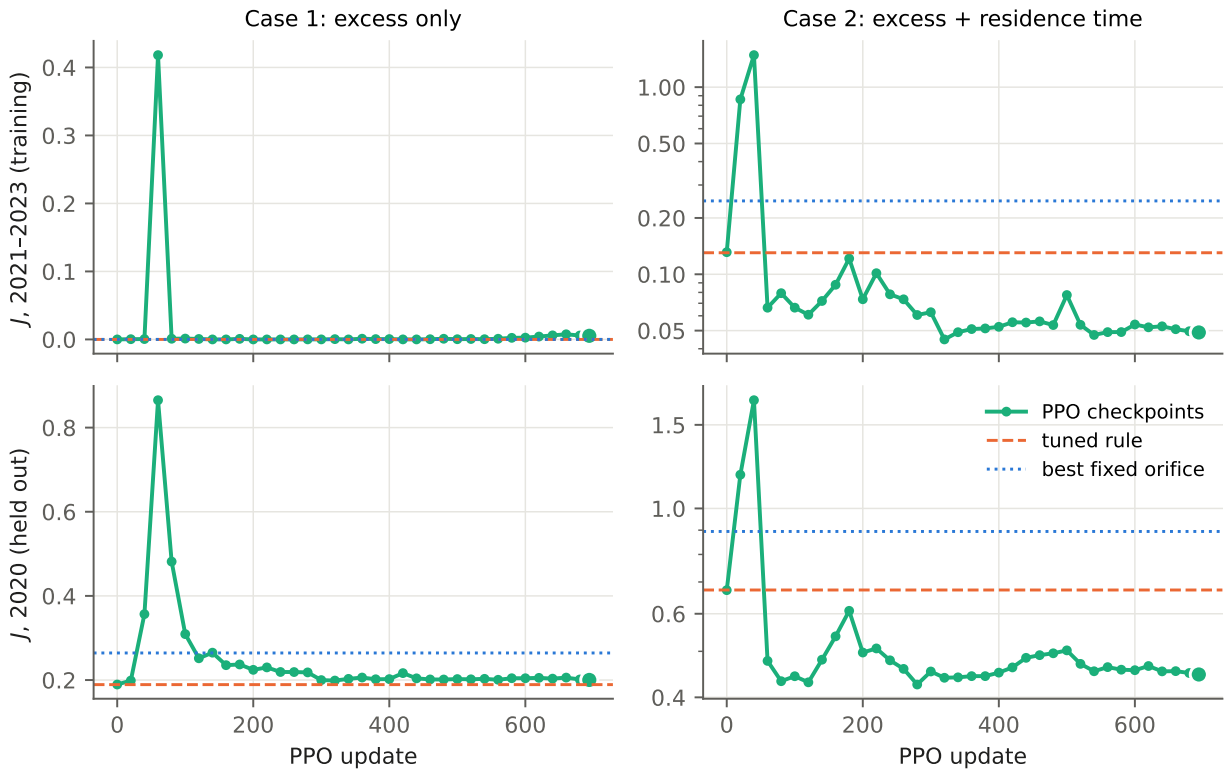


Figure 6: Held-out test. J of every PPO checkpoint (trained on 2021–2023) on the training years (top) and on 2020 (bottom), with the rule and orifice tuned on 2021–2023. The large dot is the final policy. Case 2 is on a log scale.